Automate Application Deployment

Jenkins CI/CD Deployment Automation Report

Jenkins Job	GitHub-CI-CD
Repository	https://github.com/Ar-Sawan/git-task
Branch	master
Trigger	Poll SCM
Schedule	H/5 * * * *
Pipeline	Build → Test → Deploy

1. Objective

The objective is to automate application deployment using Jenkins by configuring a build trigger, executing the CI/CD pipeline, and verifying the deployment stage and final pipeline result.

2. Build Trigger Configuration

The Jenkins job is configured with Poll SCM. The schedule is set to **H/5 * * * *, which instructs Jenkins to poll the source-control repository at approximately five-minute intervals. The Jenkins configuration screen confirms that Poll SCM is enabled.

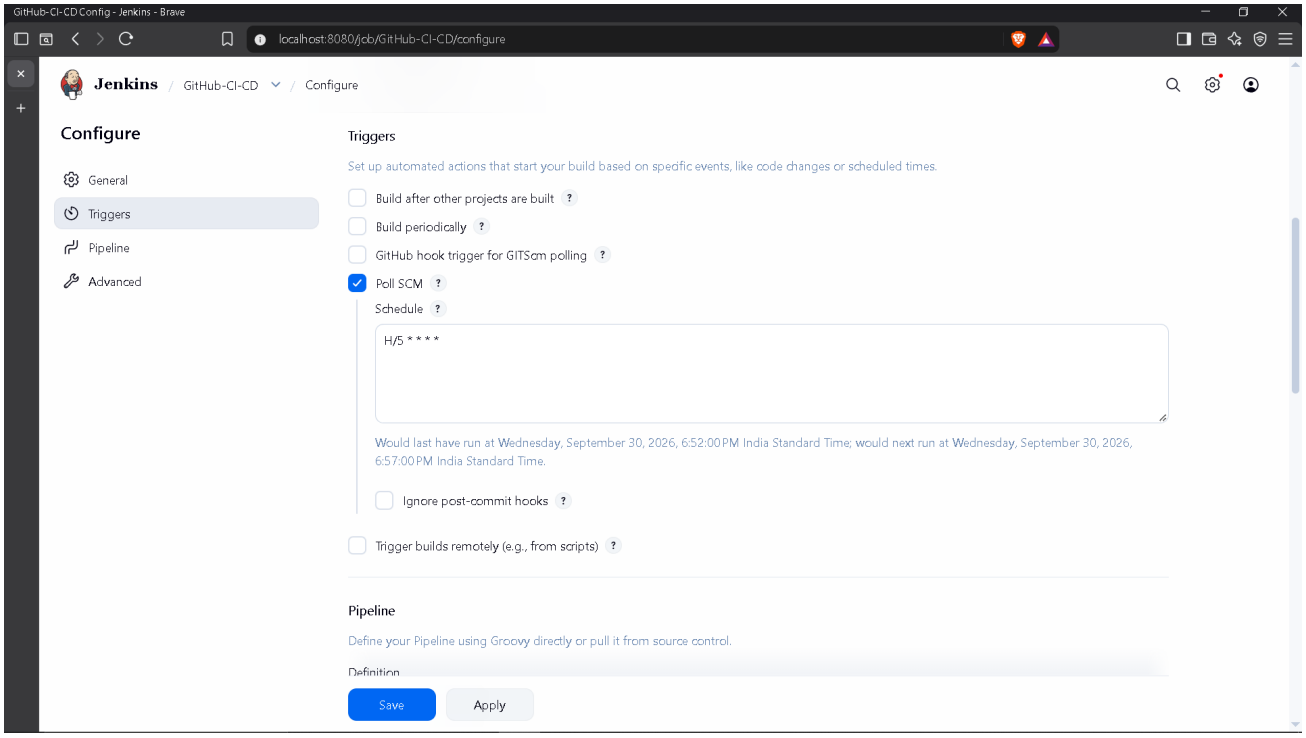


Figure 1. Jenkins Trigger configuration showing Poll SCM enabled and the H/5 * * * * schedule.

3. Deployment Pipeline Execution

The Jenkins pipeline successfully executes the configured stages. The console output shows the Build stage completing with “Build successful”, the Test stage completing with “Tests passed”, and the Deploy stage completing with “Deploy successful”. The post-build success action also reports “Pipeline completed successfully!”, followed by “Finished: SUCCESS”.

Console execution evidence

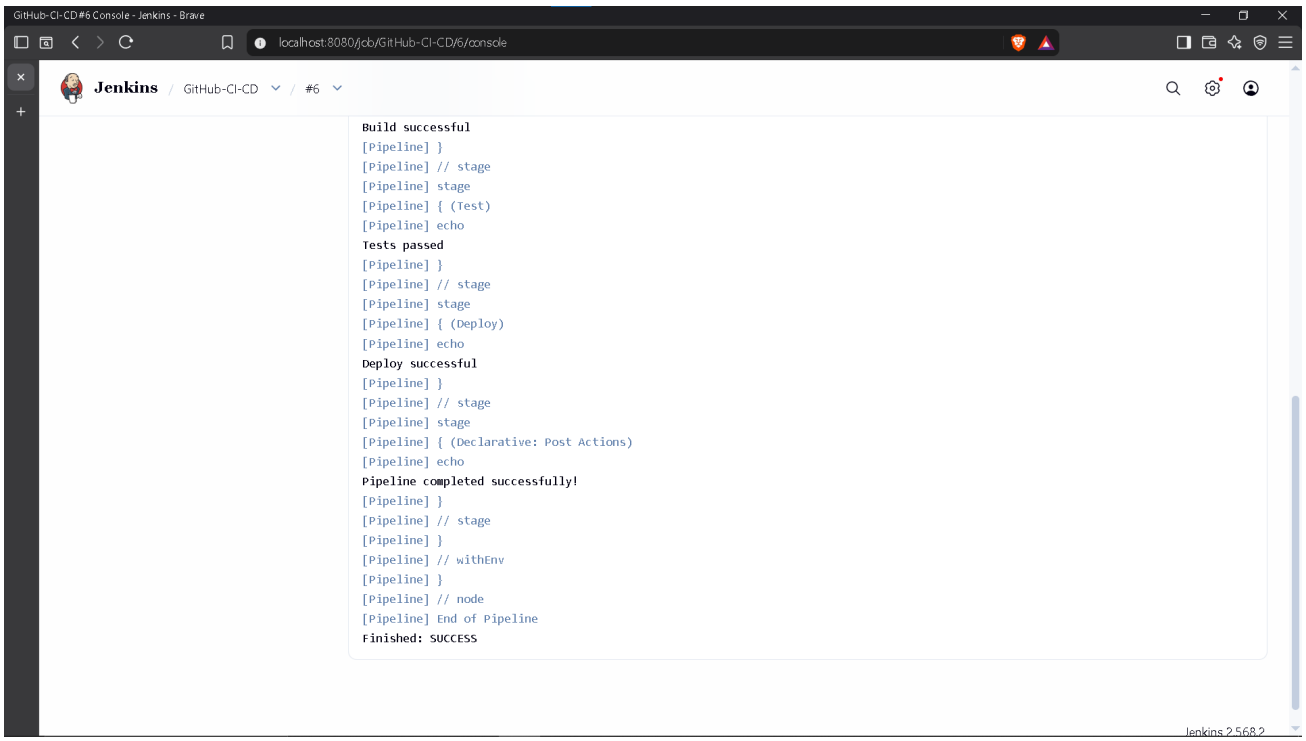


Figure 2. Jenkins Console Output showing successful Build, Test, Deploy and final SUCCESS status.

4. Automation Results

Item	Observed Result
Build trigger	Poll SCM enabled
SCM polling schedule	H/5 * * * *
Build stage	Build successful
Test stage	Tests passed
Deployment stage	Deploy successful
Post-build action	Pipeline completed successfully!
Overall Jenkins result	Finished: SUCCESS

5. Deployment Steps

• Configure Poll SCM in the Jenkins job to automatically check the GitHub repository for changes.

• Configure the GitHub repository and master branch as the pipeline source.

• Use the Jenkinsfile from source control as the pipeline definition.

• Execute Build, Test, and Deploy stages sequentially.

• Use the post-success action to confirm completion of the deployment workflow.

6. Observations

• The trigger configuration is visible and enabled in Jenkins.

• The configured polling schedule is H/5 * * * *.

• The deployment stage executes successfully after Build and Test.

• The Jenkins console reports Finished: SUCCESS.

7. Conclusion

Application deployment automation was configured and demonstrated using Jenkins. Poll SCM provides the configured source-control trigger, while the Jenkins pipeline performs the Build, Test, and Deploy sequence and reports a successful final result.